

Lesson Objective

Explain decisions about classifications of triangles into categories using variants and non-examples.

Identify shapes as triangles.

Materials: None

1. Identify the defining attributes of a triangle by counting straight sides and pointy corners. Look at this shape. How many straight sides does it have? How many pointy corners does it have?  <i>Answer: 3 straight sides; 3 pointy corners. This shape is a triangle.</i>	Show students the triangle. Point to one side and tell students that a side is one straight edge of the shape. Guide students through counting the straight sides together, touching each side as they count: 1, 2, 3. Then guide students through counting the pointy corners, touching each corner as they count: 1, 2, 3. Tell students this shape is called a triangle. A triangle has 3 straight sides and 3 pointy corners. Ask students to whisper the word triangle.
2. Determine whether shapes are triangles by checking for three straight sides, three corners, and a closed figure. Is this shape a triangle?	Present the first shape. Ask students to count the straight sides and the pointy corners. Ask if the shape is closed. Have students decide if it is a triangle and explain why.



Is this shape a triangle?



Is this shape a triangle?

Present the second shape. Have students count the sides and corners. Point out that one side is curved, not straight. Ask if it is a triangle and why not.

Present the third shape with a small gap drawn at one corner. Tell students that triangles must be closed. Ask students to imagine a pet inside the shape, like the IDL does, and ask if the pet could escape. Guide students to see the shape is open, so it is not a triangle.

For each shape, restate the rule: a triangle has 3 straight sides, 3 pointy corners, and is closed.



(Show this last shape with a small gap at one corner so it is open.)

Answer: Yes, it is a triangle (3 straight sides, 3 pointy corners, closed); No, it is not a triangle (it has a curved side); No, it is not a triangle (it is open).

3. Circle the triangles and cross out the shapes that are not triangles.

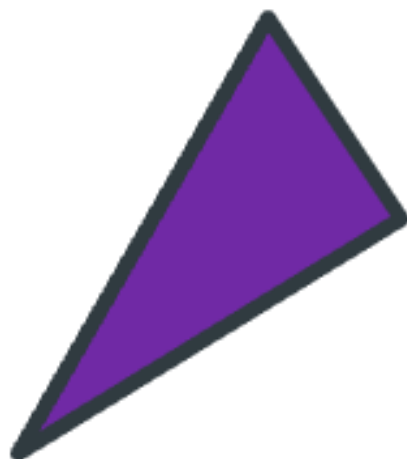
Look at the four shapes on your worksheet.

Circle the triangles. Cross out the shapes that are not triangles.



Remind students to check each shape for 3 straight sides, 3 pointy corners, and a closed figure.

Students work independently.



Answer: Circle the wide triangle and the rotated triangle. Cross out the square and the curved shape.

Monitor For

- Do students count each side and each corner exactly once, touching each as they count, without skipping or double-counting?
- Do students recognize a triangle when it is rotated or has a different shape than the first triangle they saw?
- Do students notice when a shape has a curved side instead of a straight side?
- Do students check whether a shape is closed, not just whether it has 3 sides and 3 corners?

Instructional Moves

- Have students touch each side and corner as they count aloud (1, 2, 3) to keep one-to-one correspondence. Model first, then have the student do it.
- For Problem 2, run a finger along the curved side and ask, "Is this side straight, or does it curve?" to help students name what they see.
- If a student says a shape with a gap is a triangle, use the IDL's pet language: "If a pet were inside this shape, could it escape?" Then connect: open shapes are not triangles.
- If a student says a rotated or skinny triangle is not a triangle, count the sides and corners with them and remind students that triangles can look different but always have 3 straight sides and 3 pointy corners.

Reflection Questions

- What did you count to check if a shape was a triangle? *Answer: I counted the straight sides and the pointy corners. There need to be 3 of each.*

- How did you know the shape with the curved side was not a triangle? *Answer: A triangle has all straight sides. That shape had a side that was curved, not straight.*

- How did you know the shape with the gap was not a triangle? *Answer: It was open, not closed. Triangles have to be closed.*

- Two of the triangles we looked at today looked different from each other. How did you know they were both still triangles? *Answer: They both had 3 straight sides, 3 pointy corners, and were closed.*